

# AI infrastructure and distributed systems

## Venkata Ganji

AI platform infrastructure and distributed systems engineer with 12 years in production. Across Apple, Freddie Mac and startup environments, I have shipped tooling, architected critical migrations and owned infrastructure from deployment through incident recovery.

### 01 Enterprise GenAI platform

Apple · Contract via Mphasis · Apr 2024–Aug 2026

### 02 Cyber telemetry & distributed ingestion

Cloudwick · Cyber Data Lake / Amorphic · Nov 2016–Dec 2018

### 03 Mission-critical cloud migration

Cloudwick · Client: Freddie Mac · Jan 2019–Aug 2023

### 04 Cloud automation, identity & recovery

Cloudwick · Client: Freddie Mac · Jan 2019–Aug 2023

Sanitized reconstructions and career-reported outcomes. Full case studies and supporting diagrams are available on the website.

# Enterprise GenAI platform

Apple | Contract via Mphasis | Apr 2024–Aug 2026

I owned Kubernetes, vector databases, platform operations and GPU-serving infrastructure within an internally engineered enterprise AI platform. I shipped model-weight onboarding, deployment and Solr lifecycle tooling, connecting ingestion, retrieval and model serving with production reliability.

**7.0s → 1.9s**

Approximate vector-query latency

**\$500K+**

Combined Solr + Mac infrastructure savings

**250+ nodes**

Across vector-database infrastructure

## What I owned

- Shipped and owned GPU model onboarding: Helm/Argo CD, custom vLLM images, weight staging and checksums, tensor parallelism, memory budgets and LoRA configuration.
- Authored and owned end-to-end Solr IaC and lifecycle tooling; owned operations across vector databases, Redis, MongoDB, Elasticsearch and Kafka.
- Productionized 10+ ingestion and inference microservices across Dev/UAT/Prod; owned Kubernetes operations, upgrades, routing, autoscaling and capacity planning.
- Owned observability from metrics and logs to dashboards and alerts, operational automation and recovery workflows; helped turn recurring failures into lasting fixes.

## Engineering decisions

### Separate application and model infrastructure

The architecture separates the ingestion/inference services from GPU model hosting. This lets application and model deployments have distinct capacity and runtime configurations.

### Make model artifacts part of deployment

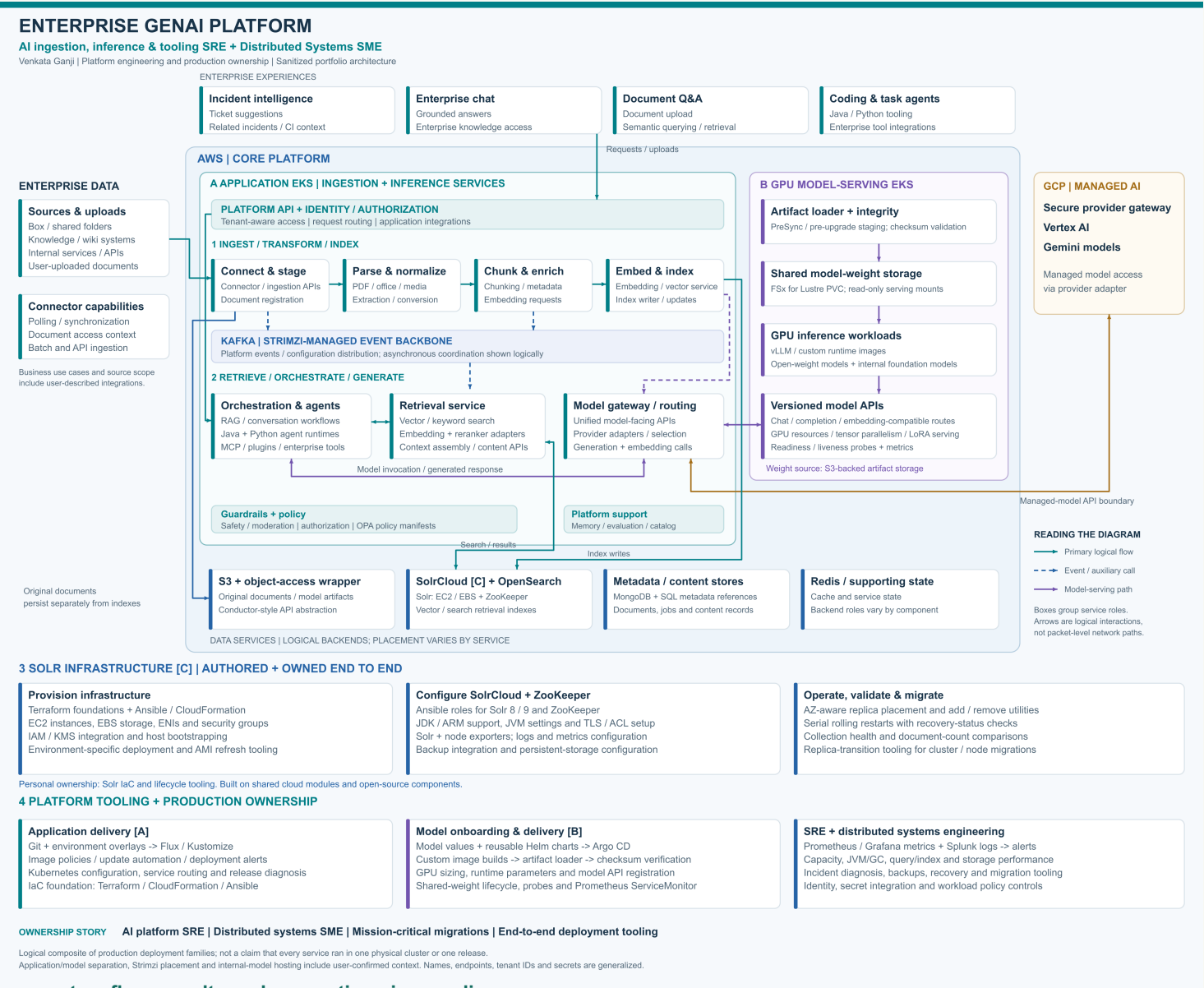
Weight staging, integrity checks, shared volumes, health probes and runtime values are first-class deployment steps. Configuration covers precision, context limits, memory budgets, tensor parallelism and LoRA serving.

### Treat retrieval as a systems problem

I worked across JVM/GC, OS settings, cache behavior, query patterns and indexes. The reported latency improvement is a vector-query result, not an LLM generation benchmark.

Scope: Sanitized logical composite of production configuration families and my account of the platform. It is not a live cluster inventory. Results are reported career outcomes; no LLM throughput, TTFT or GPU-utilization improvement is claimed. Solr authorship does not imply sole authorship of the broader platform or upstream components.

# Enterprise GenAI platform



Explore the full-size diagram, system flow, results and supporting views online →

# Cyber telemetry & distributed ingestion

Cloudwick | Cyber Data Lake / Amorphic | Nov 2016–Dec 2018

I owned the three-rack cyber data lake infrastructure end to end: hardware and network integration, operational tooling and performance tuning. I connected PCAP, NetFlow, application logs and firewall events to a correlated distributed-data platform.

## 3 racks

Capture and integration test bed

## 40 Gb/s

Packet-capture technology tuned

## 4 source types

PCAP, flow, app and firewall telemetry

### What I owned

- Implemented collection of PCAP, NetFlow, application logs and firewall events to support a correlated cyber data lake.
- Owned Supermicro servers, high-throughput NICs, IXIA and ToR / supervisor-switch integration in the three-rack capture/replay environment; tuned 40 Gb/s capture technology.
- Owned operational and performance tooling across capture, hardware and distributed data; engineered Corelight/IXIA integrations and operated Hadoop, Kafka, Spark, Solr, HBase and Elasticsearch.
- Built Prometheus monitoring for collector agents and automated cloud/on-premises deployment using CloudFormation, Ansible, Jenkins, GitLab, Kickstart, Puppet and Python APIs.

Scope: The walkthrough diagrams are simplified redraws; low-resolution previews of the supplied originals are available separately. Gateway capacities and test configurations are not promoted to production benchmarks. The separate Amorphic cloud reference is dated June 2019 and credits a team author; it is included only as product context, not as evidence that I personally designed that entire later architecture. Telemetry here does not mean OpenTelemetry.

### Engineering decisions

#### Separate functional tests from performance tests

A functionally correct collector can still be limited by disk, broker capacity or the capture path. The test environment made those dimensions visible instead of treating every failure as an application problem.

#### Keep capture and management paths distinct

The multi-rack design separates the data-transfer network from server-management access. Capture/replay and security integrations can be reasoned about independently from routine host operations.

#### Observe the collector as well as the cluster

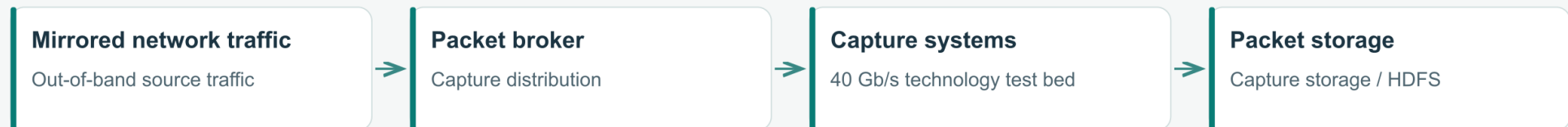
Prometheus monitoring at the collector layer complements distributed-platform operations. This gives the team visibility before data reaches the search or analytics tier.

# Cyber telemetry & distributed ingestion

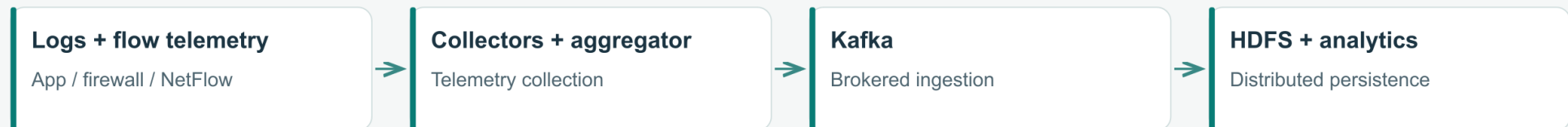
## Cyber telemetry and distributed ingestion

Logical platform roles across capture, collection, storage and investigation

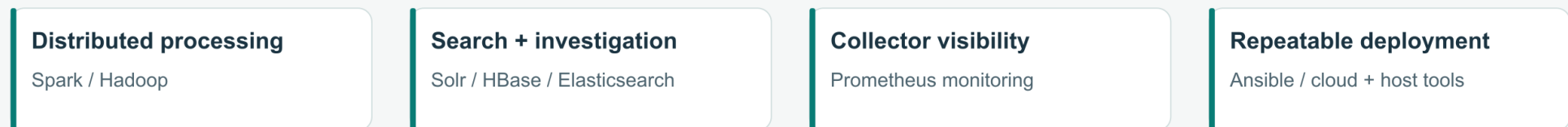
### PACKET PATH



### EVENT PATH



### CROSS-CUTTING CAPABILITIES



Rows group platform capabilities; they do not assert one pipeline for every data source.

Explore the full-size diagram, system flow, results and supporting views online →

# Mission-critical cloud migration

Cloudwick | Client: Freddie Mac | Jan 2019–Aug 2023

I delivered solutions architecture, software engineering and production operations across a nearly four-year Freddie Mac engagement. I led three mission-critical migrations, designed the Hadoop/Solr-to-S3/search transition, and led Spark migration development through production cutover.

## 3 migrations

Mission-critical programs led in the engagement

## 3–8 engineers

Approximate teams technically led

## HDFS → S3

Document-storage modernization

### What I owned

- Led three mission-critical migrations across distributed-data and live application systems during the engagement, coordinating architecture, automation, data movement, performance testing and production transition.
- Designed and productionized a document-platform migration from Cloudera Hadoop/Solr to AWS Elasticsearch/OpenSearch, ELK and S3.
- Led development of Spark migration jobs moving Solr data into Elasticsearch/ELK and HBase data into S3.
- Built reusable IaC and operations automation; designed Databricks, AWS Glue, Talend and S3 ingestion frameworks and Python/Airflow workflows; supported performance testing and production transitions.

### Engineering decisions

#### Separate transport from transformation

DistCp can address bulk transfer, while archive extraction and target naming/mapping require additional work. The design makes those responsibilities explicit.

#### Keep document and search migrations coordinated

S3 objects and search records serve different roles. The migration program covered both, so the target architecture could support application behavior as well as storage.

#### Include operations in the migration scope

Provisioning, performance testing, snapshots and recovery were part of the engineering work. A cloud cutover needed an operating model as well as a target diagram.

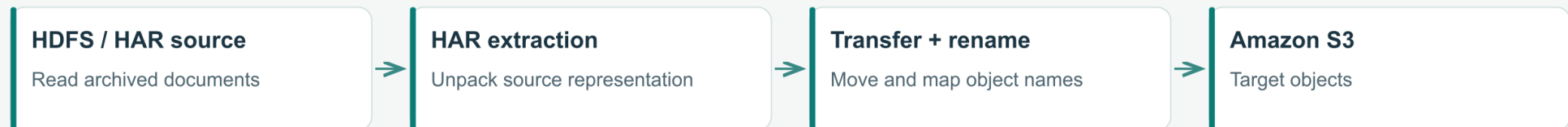
Scope: The HAR diagram presents alternatives. It does not by itself establish the selected route for every workload, total bytes migrated, zero downtime or a benchmark. The leadership and delivery claims come from my career record; the drawing explains the technical approach.

# Mission-critical cloud migration

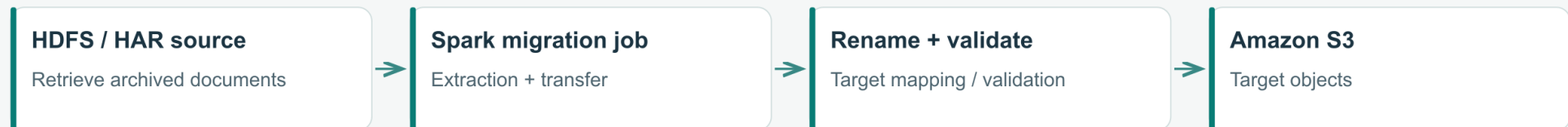
## Hadoop archives to cloud object storage

Migration approaches shown in the design; separate from the search-data migration

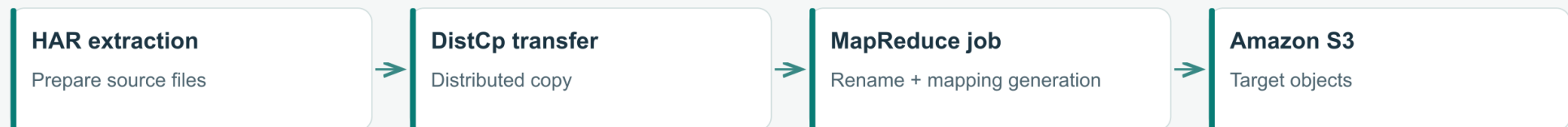
### STAGED APPROACH



### CUSTOM SPARK APPROACH



### DISTCP + MAPREDUCE APPROACH



The source drawing compares approaches; it does not establish one chosen route for every workload.

Explore the full-size diagram, system flow, results and supporting views online →

# Cloud automation, identity & recovery

Cloudwick | Client: Freddie Mac | Jan 2019–Aug 2023

I designed and productionized cloud infrastructure for a document and search platform, including proxy access, federated authentication, reusable IaC and hybrid-cloud disaster recovery. The work joined application requirements with security and operational constraints.

**~15 min**

RPO for the delivered DR environment

**~2 hours**

RTO for the delivered DR environment

**4 environments**

Federated authentication scope

## What I owned

- Designed proxy, load-balancing and autoscaling infrastructure supporting the document platform's move to AWS.
- Developed reusable Terraform, CloudFormation and Ansible provisioning/landing-zone workflows covering compute, search, networking, IAM, secrets, encryption and cross-region storage.
- Designed federated authentication using PingFederate, Vault, AWS IAM/STS, ADFS and SAML across four environments; partnered with InfoSec on on-premises-to-S3 connectivity.
- Built and productionized a hybrid-cloud DR environment using cross-region replication, S3, Airflow, Python and Elasticsearch/ELK, delivered within six months.

Scope: Drawings represent different releases and environments; the portfolio shows their relationship without implying that every identity mechanism runs in the same request path. Recovery metrics are career-reported results for the described DR environment, not a universal service guarantee.

## Engineering decisions

### Bridge the legacy application boundary

The proxy tier provides an integration point between the existing application and cloud services. The design also spreads proxies across availability zones and uses autoscaling.

### Treat identity as architecture

Directory integration, federation and temporary AWS credentials influence how applications reach the data plane. They were designed alongside network paths, not added as an afterthought.

### Make recovery and IaC reinforce each other

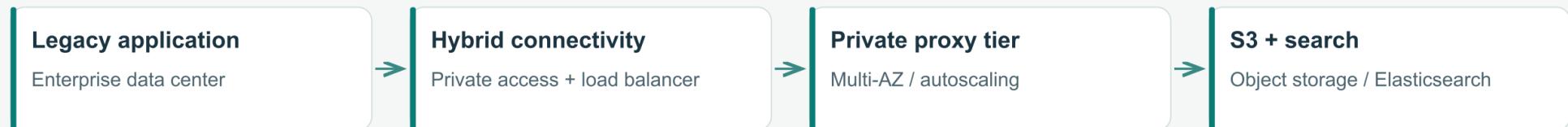
Reusable provisioning defines the target resources; backup/replication provides recoverable state. The architecture distinguishes those responsibilities, including an on-demand recovery footprint.

# Cloud automation, identity & recovery

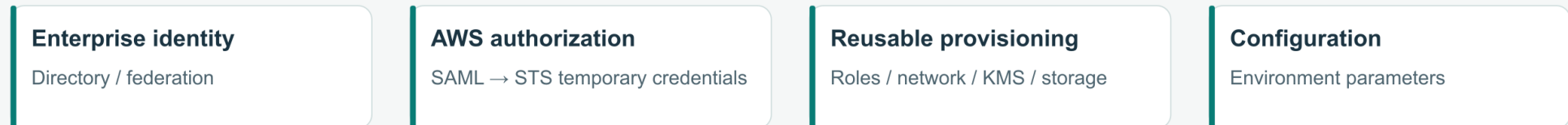
## Hybrid cloud document and search platform

Application access, cloud foundations and recovery shown as distinct responsibilities

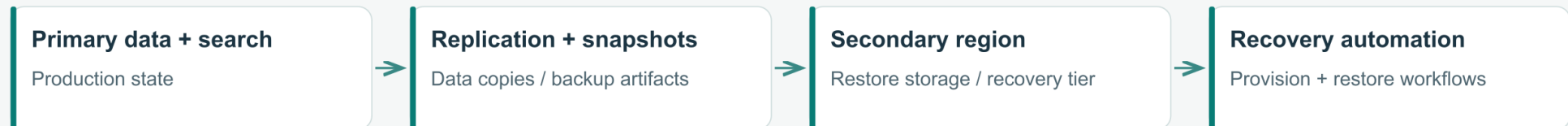
### APPLICATION DATA PATH



### IDENTITY AND INFRASTRUCTURE



### RECOVERY RESPONSIBILITIES



Logical composite of releases. DR results: approximately 15-minute RPO and 2-hour RTO for the delivered environment.

Explore the full-size diagram, system flow, results and supporting views online →